

2025 UT Invitational

10/25/2025

Dynamic Planet B/C Exam



Parameters

- Each participant may bring a 2" or smaller binder, and 2 standalone calculators of any type.
- This packet should not be taken outside the testing room! If you have this, notify the event supervisors immediately or else you will be disqualified!

Directions

- You will have **50 minutes** to complete this exam - do not begin until instructed to do so!
- All answers must be indicated on your answer document. **DO NOT WRITE ON THIS EXAM!**
- Point values are indicated for all questions & tiebreakers are indicated in test booklets. The number of tiebreakers correctly answered indicates your tie order.
- The maximum number of obtainable points on this exam is 118 points.
- You may take apart the test to split the work, but make sure to staple it back together at the end of the event.

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(any questions after the conclusion of this event may be emailed to mahima@atxscioly.org or justin@atxscioly.org)

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Exam Questions start on the next page →

Section 1 : Multiple Choice Questions

Each question in this section is worth **1 point**.

1. Sodium and chloride make up approximately what percentage of all dissolved ions in seawater by weight?
 - a. 70%
 - b. 75%
 - c. 80%
 - d. 85%
 - e. 90%
2. Name the major compound responsible for the decrease in pH in the ocean.
 - a. HCl
 - b. CO₂
 - c. CH₄
 - d. (H₃PO₄)
 - e. CO
3. The Coriolis effect causes moving objects in the Northern Hemisphere to deflect:
 - a. To the left
 - b. To the right
 - c. Upward
 - d. Downward
 - e. Straight
4. Western intensification results in:
 - a. Stronger currents on the western side of ocean basins
 - b. Stronger currents on the eastern side of ocean basins
 - c. Weaker gyres
 - d. Reduced upwelling
 - e. No significant effect
5. Ekman transport is the net movement of water at an angle of ____ to the wind.
 - a. 30°
 - b. 45°
 - c. 90°
 - d. 180°
 - e. 0°
6. The North Atlantic Gyre circulates:
 - a. Clockwise in the Northern Hemisphere
 - b. Counterclockwise in the Northern Hemisphere
 - c. Clockwise in the Southern Hemisphere
 - d. Counterclockwise in the Southern Hemisphere
 - e. There is no significant pattern of circulation
7. The main driver of surface ocean currents is:
 - a. Tides
 - b. Thermohaline Circulation
 - c. Earthquakes
 - d. Wind
 - e. Precipitation

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8. Which of the following is a warm boundary current?
- a. California Current
 - b. Gulf Stream
 - c. Canary Current
 - d. Benguela Current
 - e. Humboldt Current
9. Which of the following does not contribute to mechanical mixing in the surface layer?
- a. Surface Waves
 - b. Hurricanes
 - c. Cold Polar Air Mass
 - d. Rain
 - e. None of the above
10. Approximately, what is the current pH of the ocean?
- a. 8.0
 - b. 8.1
 - c. 8.2
 - d. 8.3
 - e. 8.4
11. Wave shoaling occurs when:
- a. Waves slow down and increase in height near shallow water
 - b. Waves break in deep water
 - c. Currents interact with tides
 - d. Waves increase in wavelength in the open ocean
 - e. Tsunamis dissipates
12. Deep-water waves travel with a speed proportional to:
- a. Wave period squared
 - b. Square root of wavelength
 - c. Water temperature
 - d. Wind speed
 - e. Wave height
13. Which of the following tides occur when the Earth, Sun, and Moon form a syzygy?
- a. spring
 - b. neap
 - c. diurnal
 - d. low
 - e. high

14. Wave refraction causes:
- a. Even distribution of energy along coasts
 - b. Dissipation of waves in deep ocean
 - c. Formation of tsunamis
 - d. Changes in salinity
 - e. Concentration of wave energy on headlands
15. Storm surge is most strongly influenced by:
- a. Tsunami wave height
 - b. Lunar phase
 - c. Salinity gradients
 - d. High winds and low atmosphere pressure
 - e. Earthquakes
16. Constructive interference of waves results in:
- a. Larger wave heights
 - b. Smaller wave heights
 - c. Wave reflection
 - d. Wave diffraction
 - e. Formation of rip currents
17. Which process contributes to the southward flow of dense, cold water in the North Atlantic, helping drive the thermohaline circulation?
- a. Western movement of ocean water caused by trade winds
 - b. Northern intensification of the subtropical gyres
 - c. Coastal upwelling in California
 - d. Deep convection in the Labrador Sea
 - e. Strengthened Antarctic Circumpolar Current
18. A Kelvin wave traveling along the equator in the Pacific Ocean always moves:
- a. Northward
 - b. Eastward
 - c. Southward
 - d. Westward
 - e. Upward
19. Which feature typically develops on the down-current side of a headland due to wave shadowing? Select all that apply.
- a. Spit
 - b. Rip current
 - c. Tombolo
 - d. Lagoon
 - e. Delta

20. The Canary Current differs from the Gulf Stream because it is:

- | | | | | |
|------------------------|---------------------------|------------------------------|------------------------|---|
| a. Warm,
deep, fast | b. Warm,
shallow, slow | c. Cold,
shallow,
slow | d. Cold,
deep, fast | e. A
thermohaline
current rather
than a surface
current |
|------------------------|---------------------------|------------------------------|------------------------|---|

21. If the Coriolis effect were suddenly “turned off,” surface ocean gyres would:

- | | | | | |
|--|--------------------------------|--------------------------------------|--|--|
| a. Reverse
circulation
direction | b. Strengthen
Ekman spirals | c. Intensify
into faster
gyres | d. Stop
circulation
and flow
linearly | e. Surface ocean
gyres won't
be effected |
|--|--------------------------------|--------------------------------------|--|--|

22. Rossby waves typically span

- | | | | | |
|--------------------|----------------------|------------------------|--------------------------|--|
| f. A few
meters | g. Tens of
meters | h. A few
kilometers | i. Tens of
kilometers | j. Hundreds to
thousands of
kilometers |
|--------------------|----------------------|------------------------|--------------------------|--|

23. Turbidites are often deposited by what type of flow?

- | | | | | |
|------------|---------------|----------------|------------|------------------------|
| a. laminar | b. frictional | c. traditional | d. density | e. All of the
above |
|------------|---------------|----------------|------------|------------------------|

24. The upper layer of oceanic crust can be characterized by what structures? Select all that apply.

- | | | | | |
|-----------------|----------------------|----------|-----------|----------------|
| a. Pillow lavas | b. Cumulate
rocks | c. Dikes | d. Gabbro | e. Troctolites |
|-----------------|----------------------|----------|-----------|----------------|

25. During an El Niño event, the thermocline in the eastern Pacific is:

- | | | | | |
|----------------------------|---------------------------------|-------------------------|---------------------------|---------------|
| a. Shallower
and weaker | b. Shallower
and
stronger | c. Deeper and
weaker | d. Deeper and
stronger | e. Unaffected |
|----------------------------|---------------------------------|-------------------------|---------------------------|---------------|

26. The Pacific Decadal Oscillation differs from ENSO mainly in:

- | | | | | |
|-------------------------------|--|------------------------------------|--|---|
| a. Strength of
trade winds | b. Timescale
and
geographic
footprint | c. Influence
of lunar
cycles | d. Dependence
on volcanic
activity | e. Absence of
sea surface
temperature
patterns |
|-------------------------------|--|------------------------------------|--|---|

27. Ocean deoxygenation is particularly severe in:
- Coral reef lagoons
 - Polar surface waters
 - Deep trenches
 - Oxygen minimum zones at mid-depths
 - Fjords
28. As the ocean absorbs more atmospheric CO₂, carbonate ion availability declines. What threshold value is most critical for reef-building corals to continue effective calcification?
- pH above 8.5
 - Temperature below 26°C
 - pH below 6.5
 - Oxygen saturation below 50%
 - Argonite saturation state (Ω_{arg}) of 3.0
29. Thermal expansion contributes roughly what fraction of observed sea-level rise?
- $\frac{1}{4}$
 - $\frac{1}{3}$
 - $\frac{1}{2}$
 - $\frac{2}{3}$
 - $\frac{3}{4}$
30. A group of scientists needs to map the shape of the seafloor to identify a potential submarine canyon. Which research tool would provide the most effective data?
- Sediment core sampler
 - CTD device
 - Sonar (multibeam or side-scan)
 - Niskin bottle water sampler
 - ADCP
31. CTDs measure conductivity as a way to measure what?
- Biodiversity
 - Current Flow
 - Underwater storms
 - Salinity
 - All of the above
32. Which water masses mix to form the “Common Water”? Select all that apply.
- NACSW
 - NADW
 - MIW
 - MBW
 - AABW

33. Which of the following are the main drivers of upwelling? Select all that apply.

- | | | | | |
|---------|--------------------|------------------|--------------------|--------------|
| a. Wind | b. Coriolis Effect | c. Lunar Gravity | d. Ekman Transport | e. Volcanoes |
|---------|--------------------|------------------|--------------------|--------------|

34. Energy balance at the surface of the ocean is a combination of which of the following? Select all that apply.

- | | | | | |
|----------------------------|---------------------------|-------------------------|---------------------------|-----------------------------|
| a. Net shortwave radiation | b. Net longwave radiation | c. Net latent heat flux | d. Net sensible heat flux | e. Net ground heat transfer |
|----------------------------|---------------------------|-------------------------|---------------------------|-----------------------------|

35. Active continental margins are characterized by _____ and relatively _____ shelves.

- | | | | | |
|------------------|----------------|--------------------|------------------|-------------------|
| a. Narrow; steep | b. Wide; Steep | c. Narrow; Shallow | d. Wide; Shallow | e. Shallow; Steep |
|------------------|----------------|--------------------|------------------|-------------------|

36. Which of the following are the most common sources of salt in the ocean? Select all that apply.

- | | | | | |
|---------|---------|---------------|-----------------|-----------------------|
| a. Land | b. Rain | c. Meteorites | d. Dead animals | e. Hydrothermal vents |
|---------|---------|---------------|-----------------|-----------------------|

37. At what fraction of a tide's wavelength would you observe the largest effect of tidal resonance?

- | | | | | |
|--------|--------|--------|--------|--------|
| a. 1/8 | b. 1/4 | c. 1/2 | d. 5/8 | e. 5/4 |
|--------|--------|--------|--------|--------|

38. A geostrophic current flows northward in the Northern Hemisphere. Where is the region of higher pressure relative to the current?

- | | | | | |
|-------------------|----------------|--------------------|----------------|---------------------|
| a. Directly ahead | b. To the west | c. Directly behind | d. To the east | e. Beneath the flow |
|-------------------|----------------|--------------------|----------------|---------------------|

39. Rossby waves exist because the Coriolis parameter (f) changes with:

- | | | | | |
|-----------------------|--------------------------|-------------|-------------|--------------|
| a. Depth of the ocean | b. Temperature gradients | c. Salinity | d. Latitude | e. Longitude |
|-----------------------|--------------------------|-------------|-------------|--------------|

40. Which statement best distinguishes Kelvin waves from Rossby waves?

- a. Kelvin waves move westward, Rossby waves move eastward
- b. Kelvin waves move eastward, Rossby waves move westward
- c. Kelvin waves and Rossby waves are the same thing.
- d. Kelvin waves move northward, Rossby waves move southward
- e. Kelvin waves move southward, Rossby waves move northward

41. Geostrophic balance is most closely responsible for the circulation of:
- a. Ocean currents
 - b. Internal waves
 - c. Atmospheric winds
 - d. A and B
 - e. A and C
42. Which process best explains why Rossby waves are so important to large-scale ocean dynamics?
- a. They dissipate energy from wind waves
 - b. They maintain tidal resonance in coastal shelves
 - c. They redistribute heat across entire ocean basins
 - d. They are not important.
 - e. They maintain tidal resonance in coastal shelves
43. Which instrument is designed to measure the speed and direction of seawater currents throughout the water column?
- a. Niskin bottle
 - b. CTD
 - c. ADCP
 - d. Sediment corer
 - e. Sonar
44. A researcher wants to study changes in ocean chemistry due to acidification over time. Which combination of tools would give the most complete data?
- a. Sonar + ADCP
 - b. Sonar + Niskin bottles
 - c. ADCP + Niskin bottles
 - d. Sonar + ADCP + Niskin bottles
 - e. None of the above.
45. In the Northern Hemisphere, a geostrophic current flows northward. On which side of the current is higher pressure located?
- a. Directly ahead
 - b. East
 - c. Directly behind
 - d. West
 - e. Beneath the flow
46. What are the different types of mesoscale eddies? Select all that apply.
- a. Dynamic
 - b. Static
 - c. Transient
 - d. Permanent
 - e. Atmospheric
47. In which month would you expect to see the most negative SAM event?
- a. January
 - b. March
 - c. May
 - d. July
 - e. November

48. In geostrophic balance, how do fluids flow?

- | | | | | |
|-------------------------------|-------------------------------|----------------------------|--------------------------------|---------------------------|
| a. Low to
high
pressure | b. High to
low
pressure | c. Along
latitude lines | d. Along
longitude
lines | e. Parallel to
isobars |
|-------------------------------|-------------------------------|----------------------------|--------------------------------|---------------------------|

Section 2 : Short Answer Questions

49. (3 pts) ADCPs have multiple frequencies by which they operate. Explain when you would use low vs. high frequencies.

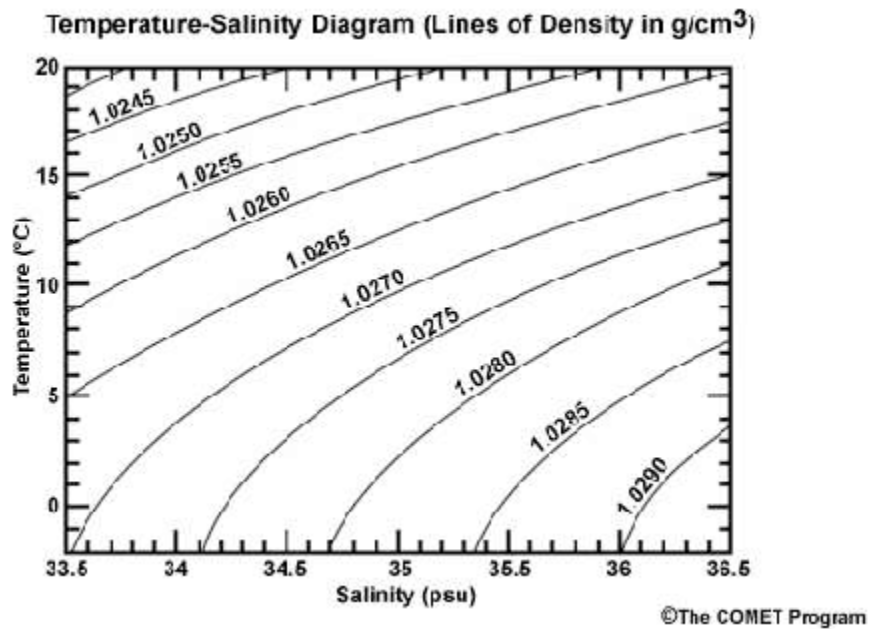


Image taken from The COMET Program “Towards a physical model of Antarctic sea ice microstructure including biogeochemical processes using the extended Theory of Porous Media”

50. (6 pts) Using the above image, answer the following:
- (1 pt) At 277K and ~34.5 PSU, what is the density of the water?
 - (2 pts) Calculate the σ_t of the density.
 - (2 pts) Why do scientists opt to use the σ_t instead of density?
 - (1 pt) What word is used to indicate lines of equal density as shown in the diagram
51. (5 pts) Answer the following questions regarding the thermocline.
- (1 pt) When can you tell you enter the thermocline of a body of water?
 - (2 pts) What role do thermoclines play in hurricane forecasting?
 - (2 pts) At mid-latitudes, what happens to the thermocline during winter vs summer?

52. (2 pts) What are the four stages in the tidal cycle in order?
53. (5 pts) Back in 1999, a program called Argo was proposed at OceanObs, an international conference of agencies who want to coordinate ocean research. The goal was to create a series of floats to help collect data and provide greater insight into the chemistry and physics of the ocean. These floats are able to control their depth, sinking and rising according to a predetermined set of instructions coded into the floats. The floats typically operate in around ten day cycles where they would drift for 9 days, sink to start a profile of various data points, and then resurface to relay their data.
- a. (3 pts) What is the mechanism behind the Argo floats' ability to sink or rise?*
 - b. (2 pts) How are Argo floats able to help scientists determine flow of ocean currents?
54. (6 pts) Answer the following questions about calculating oceanic heat transport (OHT).
- a. (2 pts) What are the three types of estimates for OHT?
 - b. (2 pts) What are the advantages to indirect estimations of OHT?
 - c. (2 pts) Why don't we use the term *temperature transport* instead of heat transport?

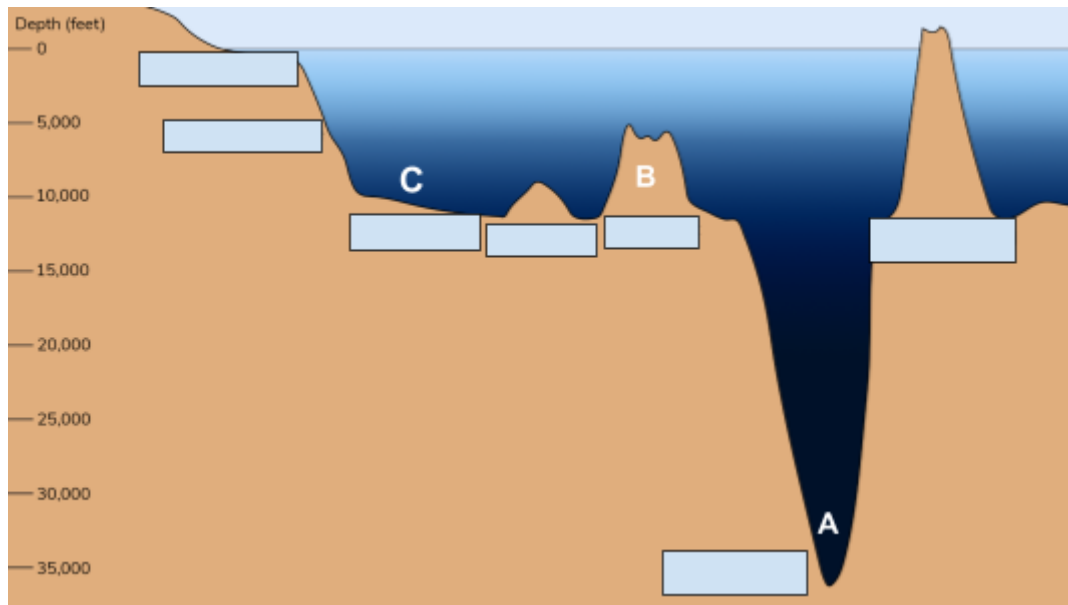
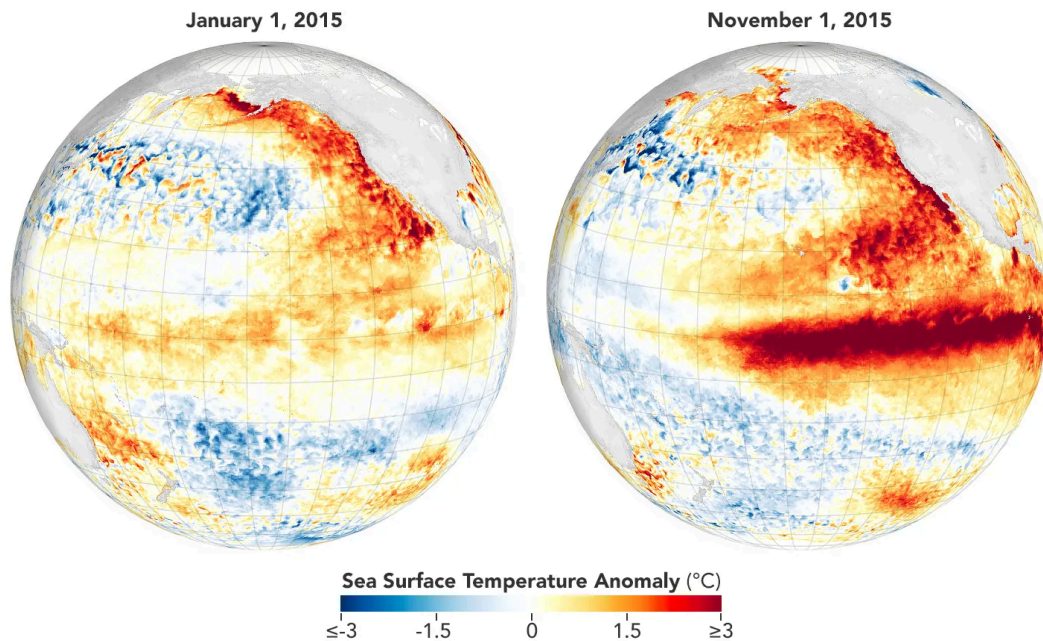


Image taken from NOAA "Ocean floor features"

55. (9 pts) Using the above image, answer the following:
- (3 pts) Identify structure A and explain how it forms.
 - (3 pts) Identify structure B and explain how it forms.
 - (3 pts) Identify structure C and explain how it is so flat.
56. (6 pts) The AMOC has been getting weaker since the 19th century due to rising temperatures attributed to climate change.
- (4 pts) Explain how and why the AMOC's strength is diminishing.
 - (2 pts) How will this affect Europe?
57. (6 pts) Waves are an extremely common phenomenon in the ocean. Scientists attempt to measure them in order to understand how they function. Let's assume that the scientists discover a wave that is 27m tall, has a wavelength of 300m, and a wave period of 10s.
- (2 pts) Calculate the steepness of the wave.
 - (1 pt) Is this wave a deep or shallow water wave?
 - (3 pts) What are the 3 things that affect wave height in wind generated waves?



*Image taken from NASA article "*Pacific Wind and Current Changes Bring Warm, Wild Weather*"

58. (7 pts) The above image shows the Pacific Ocean during neutral conditions compared to during an El Niño event. In your response:
- (2 pts) Explain the effects of an El Niño event.
 - (2 pts) Explain how equatorial Rossby waves contribute to the termination of an El Niño event.
 - (2 pts) Explain how the trade winds drive surface circulation in the equatorial Pacific under normal conditions.
 - (1 pt) Predict one ecological or climate impact of this altered circulation pattern.
59. (6 pts) Between 2022 and 2023, the Peruvian fishing industry suffered a loss of 25.43% in harvests and a 12.56% decrease in total seafood exports as a result of warming waters and political instability. Despite this, the Peruvian fishing industry is still strong with a large domestic demand for seafood. (Data is taken from *Peruvian Seafood and Aquaculture Industry Observations and Analysis* by the USDA).
- (4 pts) Why does the coast of Peru have such a rich abundance of fish?
 - (2 pt) What natural phenomenon serves as a major factor in the decrease in harvests?

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60. (9 pts) In 2011, Japan suffered from a massive tsunami that hit Tohoku, caused by a megathrust earthquake that also caused significant damage.. The economic damage was over \$300 billion and left a scar in Japan's economy. The tsunami itself ended up reaching places like Chile across the Pacific Ocean, although less intense and devastating than it was in Japan.
- (1 pt) How high on the Richter scale do you expect the earthquake that caused the tsunami to be? (Not looking for an exact number, just make an educated guess)
 - (4 pts) Explain the mechanism by which earthquakes cause tsunamis in terms of tectonics.
 - (4 pts) Tsunamis in the ocean are quite short but increase in height drastically once they approach shore. Explain how this occurs.
61. (3 pts) The Red Sea is currently expanding, slowly growing at a few centimeters per year. Located in the Middle East between the coasts of Egypt and Saudi Arabia, it suffers from high evaporation and has relatively hot waters compared to other seas. Answer the following questions regarding the Red Sea.
- (1 pt) What stage in the Wilson Cycle is the Red Sea in?
 - (1 pt) Eventually, the Red Sea will enter the next stage in its life cycle, growing into an ocean basin with continental margins. Name an ocean that is currently in that next stage.
 - (1 pt) What can you infer about the salinity of the Red Sea compared to the average salinity level based on the given information?
62. (6 pts) Answer the following questions regarding the nutrient concentrations and dissolved oxygen in the Atlantic and Pacific Oceans.
- (1 pt) Between the Atlantic and Pacific Ocean, which has the higher nutrient concentration?
 - (1 pt) Between the Atlantic and Pacific Ocean, which has the lower dissolved oxygen content?
 - (4 pts) Explain how the difference in dissolved oxygen and nutrient concentrations occur.
63. (8 pts) The Gulf Stream transports warm water from the tropical Atlantic northward toward Europe, carrying approximately 1.4 petawatts of heat energy– roughly 100 times the world's total energy consumption.
- (3 pts) Explain the mechanism by which the Gulf Stream forms and maintains its northward flow. Include the roles of wind patterns, the Coriolis effect, and density gradients.

- b. (2 pts) Western boundary currents like the Gulf Stream are narrower, deeper, and faster than eastern boundary currents like the Canary Current. Explain why.
 - c. (3 pts) Climate change could increase freshwater input into the North Atlantic through melting ice sheets and precipitation. Explain how this freshwater input would disrupt thermohaline circulation.
64. (3 pts) Answer the following questions regarding tidal patterns and astronomical influences.
- a. (1 pt) What type of tide occurs when the Moon and Sun are at 90 degrees from each other?
 - b. (1 pt) During this tidal configuration, does the tide reach its maximum or minimum range?
 - c. (1 pt) What lunar orbital position would cause this tide to have an even smaller range if it coincides with this timing?
65. (3 pts) CTD devices are essential tools in oceanographic research, deployed from research vessels to collect vertical profiles of ocean properties.
- a. (1 pt) What do the letters "CTD" stand for?
 - b. (1 pt) As a CTD device descends through the water column from surface to deep ocean, how would you expect the temperature reading to generally change?
 - c. (1 pt) CTD devices are often deployed with a rosette of Niskin bottles. What is the primary purpose of these Niskin bottles?